**Supplementary Material for: Mean correction in EEG decoding: decision geometry, temporal memory, and failure modes**

## S1. Rules that try to decide, and why ours did not.

What does not work. Two gates built on amplitude fail: matching the corrected score margins to the source session selects no correction almost everywhere, and a variance ratio that measures block-to-block drift sets the weight to zero exactly where correction is worth 0.22. A PCA concentration measure, in score space or in feature space, degenerates because the source-calibrated null saturates at 1.0 in both. A dispersion gate built on the exact identity, such as a t-statistic gate on the block displacements, reduces to the same estimate it is meant to gate. Those failures are why the rule above is continuous and estimated from the same blocks it uses, rather than a threshold applied to a single statistic. The gate experiment was also run on grasp-and-lift, but in acquisition order that dataset has no correction loss to gate, so the comparison there is not evidence either way; the numbers from the class-sorted cache are in S32.

Transductive and causal protocols must be distinguished. The batch comparisons in Table 4 include target-informed representations and use a different evaluation window; they are retained as descriptive references rather than direct competitors to the strictly causal estimators. The primary online comparison in Table 5 uses only source-only representations and evaluates all estimators on the same post-warm-up trials, starting from trial 16. On grasp-and-lift, once the trials are restored to acquisition order, that comparison gives +0.001 for the cumulative mean, +0.004 for variance-ratio shrinkage and +0.006 for predictive shrinkage (Section 4.2): no estimator loses. The losses of 0.115, 0.095 and 0.011 that an earlier draft reported for this dataset were computed on a feature cache sorted by class, are not properties of the recordings, and are withdrawn (S32).

the main text places these observations in a controlled parameter space and shows which structure of non-stationarity each failure needs.

The one-sided patch is the harmful variant, and its cost is measurable. Subtracting the estimate without adding the training mean back (8) costs 0.122 on BCI-2b, 0.047 on BCI-2a, 0.121 on SEED and 0.176 on grasp-and-lift relative to not correcting at all. That is the quantity the theory of Section 2.2 predicts, and it is an order of magnitude larger than the differences between good estimators.

Two further diagnostic figures from the development analysis - the predicted against observed change of Equation (4) and the unit medians of the source-only statistic - are not reproduced in this supplement because their source images are not part of the archived outputs; the numbers behind them are in results/final_compare.txt of the analysis archive.

## S2. The direction control

The sweep injects the origin along one direction, which is not enough: Section 2.2 says the damage is governed by the projection of the origin onto the weights, so an origin orthogonal to the weights should do nothing while raising any norm-based statistic. We ran that experiment with equal-norm origins in two directions.

**Table S1. Equal-norm origins on BCI-2b and grasp-and-lift: the norm-based statistic, the penalty on the one-sided pipeline, and the change induced by a parallel or an orthogonal displacement, by dataset and multiplier.**

| dataset | direction | norm | z_T | penalty | change |
|---|---|---|---|---|---|
| BCI-2b | none | 0 | 16.0 | −0.0443 | — |
| BCI-2b | parallel to w | 2 $\bar{\sigma}$ | 16.0 | −0.1755 | −0.1312 |
| BCI-2b | orthogonal to w | 2 $\bar{\sigma}$ | 16.1 | −0.0443 | 0.0000 |
| BCI-2b | parallel to w | 8 $\bar{\sigma}$ | 15.9 | −0.1637 | −0.1194 |
| BCI-2b | orthogonal to w | 8 $\bar{\sigma}$ | 15.6 | −0.0443 | 0.0000 |
| grasp-and-lift | none | 0 | 0.27 | −0.0147 | — |
| grasp-and-lift | parallel to w | 2 $\bar{\sigma}$ | 0.27 | −0.2279 | −0.2132 |
| grasp-and-lift | orthogonal to w | 2 $\bar{\sigma}$ | 0.27 | −0.0147 | 0.0000 |
| grasp-and-lift | parallel to w | 8 $\bar{\sigma}$ | 0.49 | −0.3225 | −0.3078 |
| grasp-and-lift | orthogonal to w | 8 $\bar{\sigma}$ | 0.50 | −0.0147 | 0.0000 |

The two directions give z_T values within 0.3 of each other on BCI-2b and within 0.01 on grasp-and-lift, which is the point: an equal-norm origin leaves the norm-based statistic essentially unchanged while the penalty moves by 0.13 to 0.31, and the orthogonal one changes the penalty by exactly 0.0000 in every cell while the parallel one changes it by −0.13 and −0.21 to −0.31. A norm-based statistic is therefore blind to the direction. The projected statistic of Equation (2) is unchanged in the orthogonal arm by construction, and it is the statistic we use.

The same decomposition on real session shifts. The sweep injects an origin; a session shift is not one. We therefore took the observed shift $d = \mu_B - \mu_A$ of each cell, set $A = P W$, and split d into the row space of A and its complement, $d_\parallel = A^{+}A\, d$ and $d_\perp = (I - A^{+}A)\, d$, so that $d = d_\parallel + d_\perp$ and $A\, d_\perp = 0$ exactly. The split has to use A and not W: ker(W) is contained in ker(P W) but is generally smaller, so a shift can move the raw scores and still leave every class-score difference untouched, and projecting on W would count it as relevant. By construction only the first can move a decision, and that is what Proposition 3 says; what the data add is that its magnitude does not order the damage: across the 845 cells the rank correlation between $\|A\, d\|$ and the penalty is +0.03, and between $\|d_\perp\|$ and the penalty +0.08, with within-dataset values between −0.04 and +0.59. The reason is the one Equation

(4) gives: the damage is a threshold-crossing function of the shift and of where the threshold sits, not a monotone function of the shift. A real shift therefore does not reproduce the clean sweep, and the paper does not claim that it does — the sweep is a controlled input, the model of Section 2.2 is what connects it to the damage, and the online comparison of the main text is what connects the model to a deployment.

## S3. A displacement model of the change, and its two parameterisations

Equations (3) and (4) make a prediction for the per-cell change once c, t and s are fixed. They can be fixed in two ways, and the two are not the same. The oracle choice takes $c = \Phi^{-1}(BA_aligned)$ and normalises by the pooled score standard deviation; it is oracle-assisted because c is read from the target's own accuracy. The model* choice takes $c = (m_1 - m_0)/(2\sigma_w)$ from the source class means and the within-class standard deviation, which is what an equal-variance Gaussian model implies.

**Table S2. Per-cell prediction of Equation (4) against the observed change, for the two parameterisations of the displacement model, one point per evaluation cell.**

| dataset | classifier | n | oracle r | oracle MAE | oracle sign (base) | model r | model MAE | BA gap |
|---|---|---|---|---|---|---|---|---|
| BCI-2b | LDA | 108 | 0.989 | 0.0130 | 0.917 (0.815) | 0.861 | 0.0567 | 0.096 |
| BCI-2b | logistic | 108 | 0.990 | 0.0127 | 0.889 (0.769) | 0.852 | 0.0578 | 0.098 |
| BCI-2b | SVM | 108 | 0.381 | 0.0224 | 0.676 (0.528) | 0.655 | 0.0203 | 0.107 |
| BCI-2a | LDA | 36 | 0.902 | 0.0320 | 0.889 (0.833) | 0.874 | 0.0414 | 0.149 |
| BCI-2a | logistic | 36 | 0.885 | 0.0330 | 0.750 (0.750) | 0.824 | 0.0563 | 0.128 |
| BCI-2a | SVM | 36 | 0.610 | 0.0161 | 0.611 (0.528) | 0.711 | 0.0217 | 0.095 |
| grasp-and-lift | LDA | 336 | 0.649 | 0.0360 | 0.693 (0.604) | 0.271 | 0.0420 | 0.213 |
| grasp-and-lift | logistic | 336 | 0.592 | 0.0364 | 0.649 (0.607) | 0.531 | 0.0355 | 0.212 |
| grasp-and-lift | SVM | 336 | 0.353 | 0.0115 | 0.420 (0.292) | 0.191 | 0.0127 | 0.248 |
| NER2015 | LDA | 320 | 0.385 | 0.0285 | 0.556 (0.406) | not estimable (282 of 320 cells have a degenerate within-class spread) | — | — |
| NER2015 | logistic | 320 | 0.371 | 0.0258 | 0.509 | not | — | — |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | (0.425) | estimable | | |
| NER2015 | SVM | 320 | 0.458 | 0.0434 | 0.666 (0.503) | not estimable | — | — |

The oracle parameterisation fits well on the BCI data (r = 0.99, MAE 0.013) and the model parameterisation fits less well (r = 0.86, MAE 0.057); the model's implied accuracy is 0.10 to 0.21 away from the observed aligned accuracy, which is a direct statement that the equal-variance assumption does not hold on these scores. Part of the oracle fit is built in, because c is read from BA_aligned and BA_aligned also enters the observed change. Equation (4) is the standard threshold-shift formula of a two-class Gaussian model; what it adds here is not a new phenomenon but the statement that the sign of the effect is carried by t and that t is a property of the training session's own calibration, which is why the same correction helps a well-calibrated LDA and hurts a margin-anchored SVM. We therefore treat it as a mechanism with a measured range of validity, not as a predictive law. The label-free check above is the evidence for the second half of that sentence.

The bound applies to the model, not to the data. The inequality $|\Delta| \le 0.1995|s|$ follows from the derivative of Equation (3). Over the 800 binary cells the prediction violates it in 0 of 800 cases and the observation violates it in 185, i.e. 23.1 per cent, with a median ratio $|\Delta_{obs}|/|s|$ of 0.29 among the violations and a maximum of 4.1: empirical balanced accuracy is a step function of the scores and can jump.

The sign, with and without a held-out half. With t* estimated on one half of the target and the sign evaluated on the other half, the sign-discrimination AUC is 0.914 [0.853, 0.958] (BCI-2b, LDA), 0.848 [0.767, 0.917] (BCI-2b, logistic), 0.848 [0.705, 0.955] (BCI-2a, LDA), 0.786 [0.618, 0.924] (BCI-2a, logistic), 0.667 [0.606, 0.725] and 0.662 [0.599, 0.721] on grasp-and-lift. The support-vector machine is unstable across settings (0.571, 0.328, 0.478) and we do not claim the model for it. Raw sign accuracy is close to the majority rate, which is why we report the AUC.

A source-only version of the same model does not work. Replacing the target-optimal threshold t* by the source-optimal one — the threshold that maximises balanced accuracy on the training scores of the fitted model, in within-class units — makes the model fully source-only. The sign accuracy it reaches is at the majority baseline: 0.815 against 0.815 on BCI-2b, 0.613 against 0.604 on grasp-and-lift, and 0.750 against 0.833 on BCI-2a, with mean absolute errors of 0.06, 0.04 and 0.05. The mechanism is therefore explanatory only, as the Discussion states.

## S4. Variants, labels, and the decision-space form

**Table S3. Alternatives to the mean correction (projection of the offset, halved correction, decision-space form) compared with the two-sided correction on the same cells.**

| dataset | none | zero | mean | proj | shrink | low5 | bal |
|---|---|---|---|---|---|---|---|
| BCI-2a | 0.5656 | 0.5254 | 0.6119 | 0.6119 | 0.5965 | 0.5873 | 0.6119 |
| BCI-2b | 0.7046 | 0.5831 | 0.7375 | 0.7375 | 0.7260 | 0.7318 | 0.7373 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| SEED | 0.4591 | 0.3377 | 0.7094 | 0.7094 | 0.5551 | 0.4460 | 0.7092 |
| NER2015 | 0.5886 | 0.5865 | 0.5909 | 0.5909 | 0.5887 | 0.5873 | 0.6039 |
| grasp-and-lift | 0.7770 | 0.7484 | 0.7815 | 0.7815 | 0.7817 | 0.7793 | 0.7815 |

The projected correction returns predictions identical to mean in 100.0 per cent of samples, as Proposition 3 requires. Halving the correction loses accuracy where it matters (0.015 on 2a, 0.012 on 2b, 0.154 on SEED), and the low-rank variant is the worst on SEED (0.446 against 0.709). The label-based correction changes nothing on the four balanced datasets (0.0000 on 2a and grasp-and-lift, −0.0002 on 2b and SEED) and gains 0.013 on NER2015, the dataset whose priors change substantially, which is what Section 2.2 predicts.

The correction composes with second-order alignment: within the Euclidean-aligned representation the mean correction still gains +0.0725 on BCI-2a and +0.0236 on BCI-2b, within the tangent representation +0.0417 and +0.0452, within CSP +0.0575 and +0.0384, and within plain log power +0.0135 and +0.0245.

For an affine model the correction can also be applied to the K scores instead of the D features, by subtracting $\delta_k = w_k'(\mu B - \mu A)$ from score k. We verified the equivalence on three linear families and three datasets: identical predictions in 100 per cent of samples, maximum discrepancy 0.0e+00, equal balanced accuracy to four decimals (0.7365 against 0.7365 for LDA on 2b; 0.7188 against 0.7188 for the SVM). The decision-space form needs the target scores and K numbers rather than the target features.

## S5. Probability calibration

**Table S4. Calibration of the corrected and uncorrected pipelines on every development cell: expected calibration error and the reliability summary.**

| dataset | n | pipeline | BA | median ECE | 95% CI | Brier | log loss |
|---|---|---|---|---|---|---|---|
| BCI-2b | 108 | none | 0.7046 | 0.0989 | [0.0786, 0.1330] | 0.3842 | 0.5690 |
| BCI-2b | 108 | zero | 0.5831 | 0.3468 | [0.3391, 0.3815] | 0.7224 | 4.7724 |
| BCI-2b | 108 | mean | 0.7375 | 0.0987 | [0.0710, 0.1086] | 0.3437 | 0.5084 |
| SEED | 45 | none | 0.4591 | 0.5467 | [0.5086, 0.5692] | 1.0781 | 14.8384 |
| SEED | 45 | zero | 0.3377 | 0.6716 | [0.6617, 0.6725] | 1.3355 | 18.4447 |
| SEED | 45 | mean | 0.7094 | 0.2404 | [0.2335, 0.3504] | 0.5769 | 7.8051 |
| grasp-and-lift | 336 | none | 0.7770 | 0.1802 | [0.1552, 0.2259] | 0.3641 | 1.1935 |
| grasp-and-lift | 336 | zero | 0.7484 | 0.2408 | [0.1941, 0.2476] | 0.4187 | 1.4418 |
| grasp-and-lift | 336 | mean | 0.7815 | 0.1671 | [0.1494, 0.2201] | 0.3545 | 1.1277 |

Because the pipelines are evaluated on the same subjects we also report the paired difference per subject. The one-sided minus uncorrected difference is +0.254 [+0.229, +0.282] on BCI-2b, +0.129 [+0.098, +0.160] on SEED and +0.030 [+0.013, +0.047] on grasp-and-lift: the one-sided pipeline is worse on all three. The two-sided minus uncorrected difference is −0.014 [−0.036, +0.006] on BCI-2b, which is a tie, −0.251 [−0.296, −0.205] on SEED and −0.006 [−0.010, −0.002] on grasp-and-lift. The correction therefore restores the calibration that the one-sided pipeline destroys on BCI-2b, improves on doing nothing on SEED, and does little on grasp-and-lift.

## S6. Collapse, and the difference between chance and collapse

The identity of Section S13 is exact by construction: on all 845 cells it agrees with the observed collapse — every prediction equal — in 124 of 124 collapsing cells and 721 of 721 non-collapsing cells. Its substantive use is that it separates collapse from chance-level accuracy. Of the 845 cells, 180 report balanced accuracy exactly 1/K after the one-sided pipeline and only 124 have collapsed; the difference is concentrated in the imbalanced dataset (NER2015: 66 cells at chance, 16 collapsed). A target whose class priors are 0.17 and 0.83 can reach a balanced accuracy of 0.5 without returning a single class.

## S7. What is left

After the correction a difference from the within-session reference remains. We call it the observed cross-protocol difference: the reference differs from the cross-session model in training-set size and in evaluation protocol, and it is not a ceiling — on BCI-2a the corrected pipeline (0.6119) is above it (0.5355), so the difference cannot be read as an information gap nor as a bound on what a label-free method could recover: 0.0198 on BCI-2b, 0.0770 on SEED, 0.0257 on NER2015 and 0.0294 on grasp-and-lift. It is not an origin effect: across all 845 cells the correlation between z_T and the residual is +0.06, and within datasets it has no consistent sign. The residual is weakly associated with the class-conditional offset h_k of Equation (5) in units of the score spread: +0.21 on BCI-2a, −0.36 on BCI-2b, +0.33 on SEED, +0.30 on NER2015, +0.55 on grasp-and-lift (pooled +0.26, four of five in the same direction). The correlation is descriptive: the class-conditional offset is computed with the target labels and so is the residual, and we do not read it as a causal decomposition. A change of the within-session covariance explains less (pooled +0.10).

## S8. Implementation details and the invariance checks

The invariances of the statistic. With the scale taken from the centred scores, z_w is invariant to (i) adding a class-common linear term to all scores, (ii) adding a constant to all scores, and (iii) translating the features in the null space of P W. We verified (i) to $4.4 \times 10^{-16}$, and (ii) and (iii) to $1.4 \times 10^{-14}$ and $2.5 \times 10^{-13}$ over 60 cells. Taking the scale from the uncentred scores instead breaks (i) and can change the statistic on a decision-equivalent classifier; that convention is not used anywhere in the paper.

The mean/dbl identity. Exact for every fit with an unpenalised intercept; verified on all 845 cells.

The z-score/none agreement. 100.0 per cent on BCI-2a, BCI-2b and grasp-and-lift with Ledoit-Wolf shrinkage, 51.8 per cent with shrinkage set to zero on grasp-and-lift, where the 256-dimensional covariance is rank-deficient. The agreement is empirical; a pipeline that relies on standardisation for equivariance should verify it.

Six properties are checked by an executable test. tests_properties.py asserts, on a real cell of BCI-2b, that mean and dbl give identical predictions sample by sample; that z-score and none do; that the feature-space and score-space corrections agree to 1e-10; that z_w is invariant to a class-common score term to 1e-12; that the collapse identity is consistent with the observed collapse; and that the causal mean at trial t is unchanged when every later trial is replaced by noise. All six pass, and the script prints one line per property.

SEED-IV. SEED-IV was converted from the distributed features during the earlier implementation. It is described here for completeness only: its numbers predate the finalised protocol and are not used as validation of any result in this paper.

## S9. A small audit of released preprocessing components

The paper's practical premise is that a pipeline can be built with a reference that is fixed at prediction time. We inspected the components of three widely used frameworks that a cross-session pipeline actually uses, and recorded which data each one takes its reference from.

**Table S5. What three released preprocessing components centre on - the MOABB cross-session example, the MNE Scaler and the Braindecode causal demean - quoted from their source code.**

| component | what it does | reference taken from | exposed at prediction time? |
|---|---|---|---|
| MOABB cross-session example [18] | CrossSessionEvaluation with CSP+LDA and tangent-space+logistic pipelines | every transform is fitted inside the training fold | no |
| MNE Scaler [mne/decoding/transformer.py] | stores a scaling estimated on the data given to fit (a fixed per-channel-type scaling by default, a fitted StandardScaler in the scalings='mean' mode) and applies it in transform | the data passed to fit, i.e. the training fold | no |

| | | | |
|---|---|---|---|
| Braindecode exponential_moving_demean [braindecode/preprocessing/preprocess.py] | causal exponential moving average, initialised from the first block of each trial | the signal up to the current sample | no, if the running window covers the session offset |

The inspected components use different sources of reference information. The MOABB and MNE examples obtain their fitted reference statistics from training data, whereas the Braindecode causal-demeaning operation examined here uses a running estimate from the observed signal. The limited audit did not identify the particular inconsistent one-sided train-test correction investigated in this study. The origin equivariance of a complete decoding pipeline nevertheless depends on how preprocessing operations are combined with model fitting and deployment, so these observations should not be read as establishing the invariance of every pipeline built from the inspected components.

## S10. An audit of the tracker

Before reading anything into the tracker column we varied its two variances and its initial covariance on the real data: the original setting (P0 = 1, process noise from the variance of successive differences, measurement noise from the current block), a matched initialisation (P0 = r), and two fixed ratios (q = 0.05 r and q = 0.50 r), all with causally estimated measurement noise. The estimator is not stable across these settings. On SEED the aggressive setting gives 0.380 against 0.459 uncorrected and the matched settings collapse to chance. The diagnostic that explains it is the one-step state error, ||state - d_b||/||d_b||, which is 0.11 on SEED and 0.99 on BCI-2a: the state is not tracking the displacement it is applied to, so the correction is applied with a vector that points elsewhere. The grasp-and-lift figures of an earlier draft (0.628 against 0.773) were produced on the class-sorted cache and are not reproduced here; they are listed in S32 with the other withdrawn values, and no claim in this paper rests on them.

## S11. What governs the damage

The displacement. For a sample mean μ and a classifier with scores S, define

$$\rho(\mu) = \| D_{\sigma}^{-1} P W \mu \| / \sqrt{K}, \quad D_{\sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_K), \quad \sigma_k = \mathrm{SD}[(S P)_{:,k}], \quad (2)$$

The scale is taken from the centred scores S P, and this matters: with σ_k taken from the raw columns, adding a class-common linear term a'x to every score leaves all predictions unchanged but changes the per-class score spreads, so the statistic would depend on how a decision function is written rather than on what it decides. With the scale from S P both numerator and scale are invariant, because P(W + 1a') = PW and (S + 1a'x) P = S P; we verify this numerically in Supplementary Section S7. Equation (2) is the definition throughout; the scalar form |w'μ_A|/σ_s used later is its binary special case, valid when K = 2 or when all σ_k are equal.

Two instances of Equation (2) are used. With $\mu = \mu_A$ it is the source-only statistic $z_w$, computable before any target data; with $\mu = \mu_B$ it is the displacement $\rho$ that the one-sided pipeline induces, computable from the target features but not from its labels.

The size. If the class-conditional scores of the aligned target were Gaussian with equal variance and the training threshold were optimal, the balanced accuracy at a threshold displaced by u standard deviations would be

$$BA(c, u) = \tfrac{1}{2} [\, \Phi(c + u) + \Phi(c - u) \,] , \qquad (3)$$

with c the standardised half-separation. At $u = 0$ this is $\Phi(c)$, and for small u it falls as $\tfrac{1}{2}\, c\, \varphi(c)\, u^2$. The coefficient $c\, \varphi(c)$ is not monotone: it increases up to $c = 1$, i.e. aligned balanced accuracy 0.84, and decreases beyond. The most exposed decoder is therefore neither the worst nor the best one.

Theorem 1 (when the correction helps). Under (3), with $c > 0$, a displacement s applied on top of a threshold error t improves balanced accuracy if and only if $|t + s| < |t|$, equivalently $s(2t + s) < 0$. The improvement region is therefore a band in (t, s): for $s > 0$ it requires $t < -s/2$, and for $s < 0$ it requires $t > -s/2 > 0$, so a correction of a given sign helps only when the existing error has the opposite sign and is at least half the displacement. This is the precise form of the statement that the sign is carried by $t^*$, and it also shows why a well-calibrated classifier ($t \approx 0$) is damaged by any displacement, while a badly calibrated one can be repaired by it.

Proof. Immediate from (3): the balanced accuracy is symmetric and decreasing in $|\, t \,|$, so $\Delta BA = BA(c, t+s) - BA(c, t) > 0$ iff $|\, t+s \,| < |\, t \,|$, which squares to $s(2t+s) < 0$. QED

The theorem is exact under its assumptions and weak when they are estimated. Taking t as the target-optimal threshold of the fitted model and s as its displacement, the sign it predicts agrees with the observed sign on 478 of the 800 binary cells, 0.598 against a majority rate of 0.564. The theorem therefore explains why the sign varies — it is a statement about the model — but it is not an estimator of the sign, and the paper does not use it as one. This is the same gap that the source-only parameterisation of the main text exposes from the other side, and it is the reason the decision rule of the main text is empirical.

The sign. Equation (3) is written around the optimal threshold, and the one-sided pipeline does not start there. Let s be the extra displacement that the one-sided pipeline applies relative to the two-sided one, and $t^*$ the displacement of the aligned threshold from the target optimum, both signed and in units of the score scale. The change is

$$\Delta = BA(c, t + s) - BA(c, t) , \qquad (4)$$

which is negative when the displacement moves the threshold away from the optimum and positive when it moves it closer. The magnitude $\rho$ is unsigned and cannot determine this sign; the sign is carried by $t^*$. Note that s and $\rho$ are different quantities: $\rho$ uses $\mu_B$ and

measures the zero-versus-none shift, while s uses μ_A and measures the zero-versus-mean shift.

## S12. What cannot be repaired

Write m_kB = m_kA + ḡ + h_k with ḡ = Σ_k π_kB (m_kB − m_kA) the prior-weighted class-conditional offset, so that Σ_k π_kB h_k = 0 by construction. Then

$$\mu_B - \mu_A = \bar{g} + \Sigma_k (\pi_{kB} - \pi_{kA})\, m_{kA} . \qquad (5)$$

The second term is the change in the mixture of the source class means; it is present even when nothing about the class-conditional distributions changes. Under the constraint the h_k do not enter the mean at all — they enter the class-conditional placement, which is where the residual of the main text comes from. Subtracting μB − μA removes the class-conditional offset and the prior-induced mixture term together; on a balanced design the second term vanishes, and when the priors change it does not.

If the class-conditional distributions are unchanged, p_B(x | y) = p_A(x | y), and the classes are distinguishable in the score space (the mixture components are not degenerate), the target priors are identifiable from unlabelled target data and can be estimated by the EM procedure of Saerens et al. [12]. Without that assumption a prior change and a class-conditional offset are confounded, since both act on the first two moments of the mixture, and no label-free method separates them.

## S13. The collapse identity

After zero centring the classifier returns a single class c for every target sample of the finite target set X_B if and only if, for every k ≠ c,

$$w_k'\mu_B - w_c'\mu_B \;\geq\; \max_{x \in X_B} [\, s_k(x) - s_c(x) \,] , \qquad (6)$$

with equality resolved by the argmax tie-breaking rule (the smallest class index wins in our implementation). This is an identity: it rewrites argmax_k s_k(x − μB) = c for all x in X_B. The maximum is taken over the finite target set, not over RD; over RD it need not be bounded when the two weight vectors differ. It is useful operationally, because it can be checked from the fitted weights, the target mean and the target scores before any label, and because it distinguishes collapse from chance-level accuracy. It is not a predictive method, and the simplest label-free check of collapse remains counting the distinct predicted classes.

## S14. The origin sweep, as a check on the implementation.

none and mean are exactly invariant: across 845 cells and ten settings the largest absolute change in balanced accuracy was 0.0e+00 for both, in every dataset (Fig. 2c). This is the numerical confirmation of Proposition 1, not an independent discovery: the algebra says the number must be zero, and the experiment says the implementation behaves as the algebra says. The one-sided pipeline decays toward 1/K as z_T grows, but it does not reach it uniformly within the sweep: on grasp-and-lift the fraction of cells at chance rises from 0.00 to 1.00 as z_T goes from 0.26 to 54.9; on BCI-2a from 0.08 to 0.97; on NER2015 from 0.05 to

0.93; on BCI-2b the sweep starts at z_T = 26 where 0.60 of the cells are already at chance and ends at 0.78; on SEED it starts at z_T = 34 where 0.90 are already at chance and ends at 1.00.

## S15. A source-level statistic, and its boundaries

**Table S6. The source-only statistic at the level of dataset-by-representation units: median, range and the penalty of the one-sided pipeline.**

| dataset | representation | n | z_T | z_w | penalty | gain |
|---|---|---|---|---|---|---|
| BCI-2b | csp | 27 | 0.22 | 0.068 | +0.018 | +0.026 |
| NER2015 | time mean | 320 | 0.20 | 0.230 | 0.000 | 0.000 |
| grasp-and-lift | log power | 336 | 0.25 | 0.289 | −0.015 | 0.000 |
| BCI-2a | csp | 9 | 0.40 | 0.287 | +0.024 | +0.059 |
| BCI-2a | tan † | 9 | 0.37 | 0.309 | +0.014 | +0.031 |
| BCI-2a | euc_ea † | 9 | 0.64 | 0.597 | −0.014 | +0.059 |
| BCI-2a | euc | 9 | 7.87 | 4.399 | −0.184 | +0.017 |
| BCI-2b | tan † | 27 | 21.2 | 5.117 | −0.160 | +0.020 |
| BCI-2b | euc_ea † | 27 | 14.2 | 5.405 | −0.137 | +0.018 |
| SEED | differential entropy | 45 | 34.2 | 12.36 | −0.029 | +0.211 |
| BCI-2b | euc | 27 | 57.1 | 23.31 | −0.186 | +0.021 |

On the source-only representations — 773 of the 845 development cells, seven units, since euc_ea and tan are target-informed — the rank correlation between z_w and the penalty is −0.857, with a cluster bootstrap over datasets of [−1.000, −0.585] (median −0.862, 4,993 resamples), and the direction is the same at the cell level (AUC 0.831 for a loss larger than 0.05). Including the two target-informed representations in the unit-level analysis changes the correlation only to −0.827, so the statistic is not being carried by the representations a source-only analysis should not use. The statistic also resolves the four representations of BCI-2a, which the norm-based z_T does not: euc has z_w = 4.40 and a penalty of −0.184, while csp, tan and euc_ea sit between 0.29 and 0.60 with penalties between +0.024 and −0.014.

The statistic is defined relative to a feature origin. z_w uses the absolute source mean, so it answers the question "how far is this family's origin from the classifier's reference, in units of the score spread", not "how exposed is this family in an absolute sense". Translating every feature by a vector c changes the numerator to $\|D_{\sigma^{-1}} P W(\mu_A + c)\|$; a translation in the null space of PW leaves it unchanged, and a translation along the weights moves it one-for-one. The numbers in Table 6 are therefore comparable within one convention — the convention in which each dataset's features are built — and not across conventions; a feature family that is shifted by a constant before the pipeline would receive a different z_w. This is one reason we do not offer an absolute threshold, and it is a limitation of the quantity rather than of the measurement.

Where it fails. At the dataset level the relationship is substantially less reliable. Across the five development datasets and using source-only representations, the dataset-level

Spearman rank correlation is −0.80. With only five datasets this is descriptive and does not establish a transferable operating threshold.

A threshold of 0.354 selected by Youden's J on the development data is retained only as an exploratory, within-development diagnostic. In leave-one-dataset-out evaluation the operating rule fails on SEED (sensitivity 1.000, specificity 0.000) and NER2015 (sensitivity 0.273, specificity 0.864); the corresponding values for BCI-2a, BCI-2b and grasp-and-lift are 0.714/0.864, 1.000/0.574 and 0.704/0.828.

The source-only statistic can therefore describe differences between feature families, but it does not provide a reliable rule for deciding whether to correct an unseen session. Results from the earlier SEED-IV implementation are excluded from this evaluation.

**Table S7. Leave-one-dataset-out operating point of the source-only statistic: the threshold chosen on the remaining datasets, and the sensitivity and specificity it achieves.**

| held out | n | threshold | sensitivity | specificity | precision |
|---|---|---|---|---|---|
| BCI-2a | 36 | 0.354 | 0.714 | 0.864 | 0.769 |
| BCI-2b | 108 | 0.354 | 1.000 | 0.574 | 0.753 |
| SEED | 45 | 0.354 | 1.000 | 0.000 | 0.489 |
| NER2015 | 320 | 0.475 | 0.273 | 0.864 | 0.188 |
| grasp-and-lift | 336 | 0.369 | 0.704 | 0.828 | 0.627 |

The conclusion we draw is that a source-level statistic can rank feature families within a set of datasets and cannot decide which session, subject or representation will fail. We report it because it is the natural quantity to compute before a target session exists, and because its three failures are as informative as its success.

## S16. Visual synthesis of causal deployment and failure regimes

Figs. S1-S3 provide schematic summaries of the preceding derivations and experiments. They are explanatory illustrations rather than additional EEG measurements or independent validation of the proposed failure mechanisms.

The causal protocol separates training-only reference information from the sequential target stream. The first 15 target trials provide a warm-up block; prediction starts at trial 16 and accesses only completed historical blocks (Fig. S1).

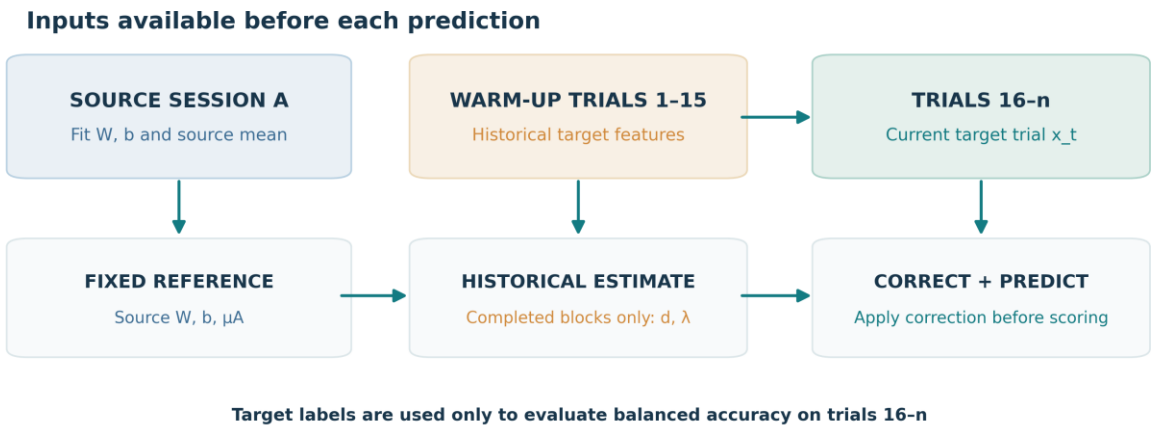



Fig. S1. Schematic of the primary strictly causal evaluation protocol. The source classifier and its reference statistics are fixed before target deployment. Trials 1–15 form a non-scored warm-up block, and post-warm-up predictions use completed historical blocks only; target labels are used for evaluation, not adaptation. Conceptual illustration prepared with AI-assisted drafting (OpenAI ChatGPT, GPT-6) and rendered using Matplotlib 3.10.8; no experimental observations are depicted.

Proposition 3 concerns score differences rather than the norm of an arbitrary feature shift. The two-dimensional conceptual example in Fig. S2 contrasts displacements orthogonal to the decision-relevant direction with displacements that change class-score margins.




Fig. S2. Decision-space geometry underlying Proposition 3 (conceptual illustration, not measured EEG data). A displacement orthogonal to the classifier decision direction leaves pairwise margins unchanged (a), whereas a displacement with a nonzero projection onto the centred classifier weights changes margins (b). Open markers indicate schematically displaced positions. Prepared with AI-assisted drafting (OpenAI ChatGPT, GPT-6) and rendered using Matplotlib 3.10.8.

The failure-regime summary in Fig. S3 separates the causal significance of stable additive offsets, within-session drifts, abrupt changes and class-prior changes. The categories describe controlled simulation conditions, not causes identified in any particular real EEG dataset.

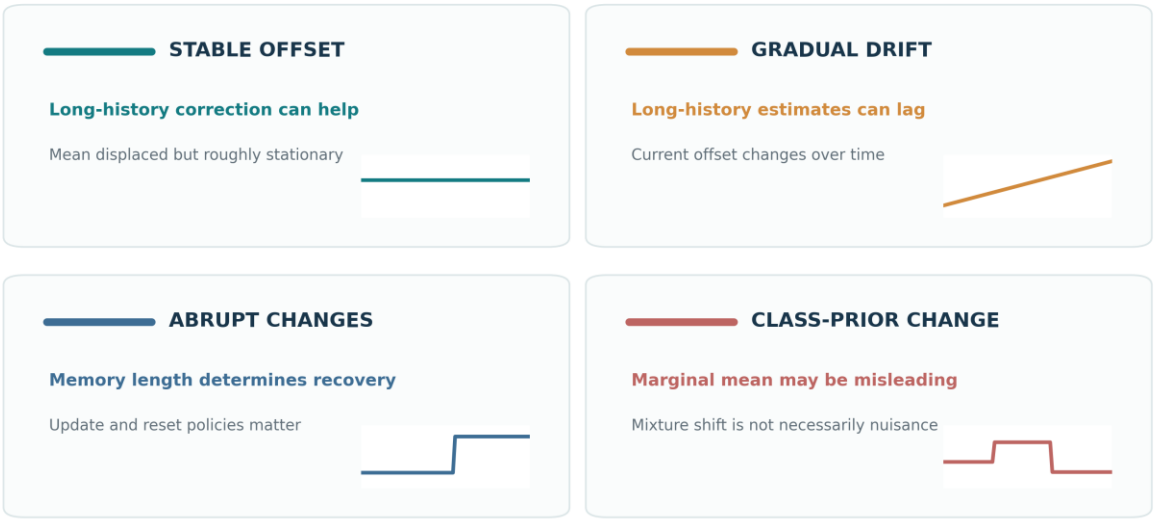



Fig. S3. Qualitative summary of four synthetic failure regimes. Stable offsets, gradual drift, step changes and class-prior variation can change the relative benefit of mean correction; no universally superior memory length or correction weight is implied. Miniature traces are illustrative only and do not represent measured EEG. Conceptual illustration prepared with AI-assisted drafting (OpenAI ChatGPT, GPT-6) and rendered using Matplotlib 3.10.8.

## S17. The verification procedure

The results reduce to a procedure with three measurements, one correction and one test.

Measure the origin distance of the feature family. The statistic z_w in Equation (2) is computed from the fitted classifier, the source feature mean and the centred score spread. In the observed development representation units, one-sided correction penalties range from +0.024 to −0.015 when z_w is below 0.35, and from about −0.03 to −0.19 when z_w exceeds 4. These ranges describe the evaluated feature conventions and must not be read as universal operating thresholds. The inconsistent leave-one-dataset-out screening results further indicate that the statistic cannot reliably decide whether correction should be enabled for an unseen recording.

Check the collapse identity before believing an accuracy. If all target predictions are equal the classifier has collapsed, and its balanced accuracy is 1/K by identity.

The invariance test. Translate every feature by a constant, refit, and check that the predictions are unchanged. The test is one line of code and needs no labels. It has two blind spots that follow from Proposition 3: a translation in the null space of P W changes nothing,

and a target sample that does not cross the moved boundary changes no label. The test is therefore a sufficient but not a necessary detector, and it should be run with a translation large relative to the score range; we use ten within-session standard deviations in the sweep of the main text.

Report the residual separately. The correction recovers the origin component; the rest is an observed difference from the within-session reference, not an identified information gap, and reporting the two separately prevents a corrected pipeline from being credited with the whole difference.

## S18. What the verification establishes, and what it does not

Three kinds of statement are mixed in a paper of this shape, and only the second is specific to EEG. The first is a corollary of linear discriminant theory: a constant added to a feature vector moves the intercept, the prediction-relevant content of a reference is its projection onto the weights, and score differences are unchanged by class-constant shifts. We prove these and do not claim them as discoveries. The second is an empirical regularity of cross-session EEG features: which feature families carry a free origin, how large the resulting damage is on five public datasets, how the damage depends on the classifier's threshold error, and in what ways a source-level statistic fails. The third is a change in practice: centre both sides, run the invariance test, and do not use a norm-based origin statistic. The paper's contribution is the second and the third; the first is the scaffolding, and we say so because a reader who expects a new algorithm should know what this is not.

## S19. Why some features are exposed and others are not

z_T is large when the feature carries a free additive constant. Log power has three: the logarithm of a reference power, a floor added before the logarithm, and a change of the reference electrode that adds an offset to the potential before a non-linear transform. Differential entropy inherits all three. A difference of two conditions measured within the same trial has none, which is why the time-domain contrast features of NER2015 have z_T = 0.20 and a penalty of 0.000. This is a statement about measurement conventions, not about neurophysiology: the origin is set by the amplifier, the montage and the code, and none of the three is reported in a typical methods section.

The electrical reading should not be pushed far. An amplifier gain change and a reference-electrode change both alter the covariance of the raw signal, and a reference change is not a pure translation of the feature vector; what is additive after a log transform is the magnitude of a gain change, not every consequence of a montage change.

## S20. Relation to alignment methods

Recent work has attributed part of the cross-session benefit of a tangent-space pipeline to test-time recentering [20]. This motivates examining the availability of recentering under causal deployment rather than assuming access to the complete target session. Our primary causal evaluation uses source-only representations and begins after a 15-trial warm-up. Under this protocol mean correction improves balanced accuracy on the motor-imagery

benchmarks and SEED, and leaves grasp-and-lift unchanged once the trials are in acquisition order (+0.001); the negative transfer this section once reported for grasp-and-lift came from a cache sorted by class and is withdrawn (S32). The point that survives is that recentering availability, correction strength and target-session performance are separate considerations.

## S21. Temporal structure of the shift covaries with the loss

This section once reported that the temporal structure of the block displacement covaries with the correction loss on grasp-and-lift (rho = +0.776), together with the prior/conditional decomposition of the block mean and the long-short interaction. Every number in it was computed on the feature cache sorted by class, so it describes the artefact rather than the recordings. The material is reproduced in S32 for the audit record and is not evidence about real recordings; in acquisition order grasp-and-lift shows no correction loss to correlate with (+0.001, Section 4.2).

## S22. The independent replication on sleep-EDF: protocol, audit and numbers

Protocol. The comparison in Section 4.7 was fixed before the data were read (INDEPENDENT_TEST_PROTOCOL.md). One continuous night is cut into consecutive 30-minute series of 60 epochs of 30 s; every ordered pair of series within the same night is a cell, with the earlier series as source and the later one as target. Features are the log-variance in four bands (0.5-4, 4-8, 8-12 and 12-30 Hz) of two EEG channels (Fpz-Cz, Pz-Oz) per epoch. A linear discriminant with lsqr and automatic shrinkage is fitted on the source series only. The correction is the mean of the completed blocks of 15 epochs, the first block is unscored, and full strength (lambda = 1) is compared with no correction (lambda = 0) on the same trials, scored by balanced accuracy.

Leakage audit. Four properties were checked. (i) Fit isolation: the classifier, the features and the covariance see the source series only; no target statistic enters feature construction, standardisation or model selection. (ii) History: the mean used at block k is the mean of blocks 0 to k-1, so no trial from the scored block or from any later trial enters the correction. (iii) Warm-up: the first block is excluded from scoring, so the cold-start estimate affects no number. (iv) Class imbalance: scoring is balanced accuracy throughout; the classifier uses the source class priors without re-weighting, which is a protocol choice rather than a channel for leakage.

One caveat, stated as such. The re-run filters each 30-minute series separately, as the protocol specifies, so no target epoch sees samples from outside its own series. In the first implementation the 0.5-30 Hz band-pass was applied to the whole night, which is non-causal: a target epoch's feature then used samples recorded after it. The direction of that bias is not guaranteed for every cell - filtering changes the source and target features, their autocorrelation and the signal near class boundaries - so the non-causal filter is one reason the protocol-consistent re-run was done, not an argument that the first numbers were safe.

Numbers. Forty-five recordings from 32 subjects give 4,983 cells. At the subject level, which is the primary unit and is the equal-weight mean of each subject's night means (13 subjects contribute two recordings): mean difference -0.1023, bootstrap interval [-0.114, -0.090], negative in 32 of 32 subjects, sign test p = 4.7e-10. Pooling the cells instead gives -0.1017. At the recording level, reported as secondary: -0.1018, [-0.113, -0.091], negative in 45 of 45, p = 5.7e-14.

Bookkeeping. The subject identifier is the first five characters of the file name (SC4xx); the sixth is the recording index. A four-character key collapses ten subjects into one and returns seven units instead of 32. An earlier version of the aggregation script used the four-character key; it was corrected before the numbers reported here were produced, and re-aggregating the stored cells with the five-character key reproduces them. The corrected script and both sets of cells (4,983 for the re-run, 5,537 for the first implementation) are in the analysis repository (sleep_edf/).

Protocol-consistent re-run. The first implementation of this test used six stage labels (N3 and N4 separate), filtered the whole night with an MNE FIR band-pass before the band variances, allowed a series to span unlabelled epochs, admitted nights with fewer than four series, and pooled cells within a subject. The re-run reported here uses the frozen five stages, a fourth-order Butterworth zero-phase filter applied to each 30-minute series separately, contiguous labelled runs only, the frozen night-level inclusion rules and equal-weight night means. Both runs are archived; the first gave -0.1084 over 5,537 cells with 32 of 32 subjects negative. Every difference, including the cohort size (45 nights from 32 subjects rather than 20 subjects with two nights each) and the pairwise two-class requirement, is listed in INDEPENDENT_TEST_DEVIATIONS.md, which the paper cites and which ships with the submission.

Single-stage series. Of 42,616 candidate pairs, 28,124 are excluded because the source series contains one stage only, which LDA cannot be fitted to, and 9,509 because the target series contains one stage only, where balanced accuracy degenerates to the recall of that stage. The second rule is an evaluation choice, not a computational one: removing it leaves 14,492 cells and a subject-level mean of -0.152 [-0.180, -0.124], negative in 32 of 32 subjects (p = 4.7e-10), and a recording-level mean of -0.140 with 42 of 45 recordings negative. The sign of the effect does not depend on that rule. Both runs are archived in the repository.

Class-set overlap. The effect does not depend on the target asking about classes the source has seen. Stratifying the 4,983 cells by the relation between the two class sets gives: identical sets, 693 cells, subject mean -0.061, 29 of 32 subjects negative (sign test p = 2.6e-6); target a subset of the source, 1,070 cells, -0.147, 31 of 32; target containing a class absent from the source, 3,220 cells, -0.092, 32 of 32. The sign survives in the cleanest subset, where the classifier is asked only about classes it has seen; the magnitude is smaller there, so part of the effect size, but none of its direction, depends on unseen classes.

Strict causality. The primary run filters each 30-minute series with a zero-phase filter, so a feature can depend on later samples in its own series. Repeating the whole analysis with

forward-only filters - the 0.5-30 Hz band-pass and the four band filters applied once, with the filter state carried across the night, so that no sample ever sees a later one - gives a subject-level difference of -0.1017 ([-0.1135, -0.0894]), negative in all 32 subjects (sign test p = 4.7e-10) and in all 45 recordings. The negative transfer therefore does not depend on the non-causal filter; the causal claim still applies to the correction step, and the paper does not claim an end-to-end real-time system.

## S23. Estimator details moved from the main text

The predictive weight. With d_b the decision-space displacement of completed block b and d_bar_{-b} the mean of the other completed blocks, the weight is lambda-hat = clip[0,1] of sum_b <d_b, d_bar_{-b}> divided by sum_b |d_bar_{-b}|^2, with the sum over b = 2 ... B. Below three completed blocks the ratio is undefined and the weight falls back to B/(B+3); a denominator below 1e-15 gives zero. The numerator is uncentred, so the criterion asks whether a block agrees with the mean of the others rather than whether it is close to their level.

Cold start. The first block of m = 15 target trials is warm-up: it is not corrected and not scored, and evaluation starts at trial m + 1. For the first evaluated block only one block is complete, so between-block dispersion cannot be estimated; with u_i = P W x_i the centred score vector of a warm-up trial and u_bar its within-block mean, the variance-ratio weight becomes lambda_1 = |d_1|^2 / (|d_1|^2 + (K/m^2) sum_i |u_i - u_bar|^2). The within-block noise term therefore carries exactly one factor of K, as does the denominator K V_d / B of Eq. (9) in the main text.

Why these are not in the main text. The main text states the two estimators that the paper's claims depend on (completed-block mean and the variance-ratio weight of Eq. 9) and reports all six in Tables 4 and 5. The predictive rule is a diagnostic that, on the acquisition-order grasp-and-lift cache, is within 0.006 of no correction; its derivation is kept here for completeness.

## S24. Estimator and strength matrix on sleep-EDF

Table S24. Subject-level balanced accuracy on sleep-EDF, 4,983 cells, 45 recordings, 32 subjects. Each cell is a causal estimator with the first block of 15 epochs unscored; lambda = 0 is no correction.

estimator | lambda = 0 | 0.25 | 0.50 | 0.75 | 1 | delta at lambda = 1 | subjects negative

cumulative | 0.4415 | 0.4370 | 0.4150 | 0.3755 | 0.3345 | -0.1070 | 32 of 32

completed block | 0.4415 | 0.4384 | 0.4176 | 0.3793 | 0.3391 | -0.1023 | 32 of 32

sliding window 50 | 0.4415 | 0.4369 | 0.4149 | 0.3751 | 0.3336 | -0.1078 | 32 of 32

exponential 0.05 | 0.4415 | 0.4387 | 0.4304 | 0.4125 | 0.3897 | -0.0517 | 32 of 32

variance ratio | 0.4415 | 0.4396 | 0.4236 | 0.3928 | 0.3595 | -0.0820 | 32 of 32

The loss is therefore a property of the correction, not of one estimator: every rule loses at full strength and every rule loses in every subject. The magnitude does depend on the estimator - the exponential average, the shortest memory here, loses least (-0.052) and the two block-scale rules lose most (-0.102 to -0.108) - but no rule is safe, which is the claim the main text makes. The same matrix on the grasp-and-lift cache in acquisition order gives deltas between +0.001 and +0.006, i.e. no loss for any rule.

Matched scored trials. In the ordering experiment the first block is unscored, so the two orderings do not score exactly the same trials. Restricting both to the trials scored under either ordering (the intersection of the two scored sets) gives -0.0014 in acquisition order and -0.1145 in the class-sorted order on the identical trial set, a difference of -0.1131 with the sorted order lower in 292 of 336 cells. The ordering effect is therefore not produced by a different warm-up sample.

## S25. Temporal structure of the block-mean displacement

Table S25. Median over cells of four statistics of the block-mean displacement, block length 15. The grasp-and-lift row was computed on the class-sorted cache and is superseded by the acquisition-order rebuild of Section 4.5; it is kept here only so that the earlier diagnostic can be traced.

dataset | R-squared of a linear ramp | trend | block-to-block change | blocks per session

grasp-and-lift (sorted cache) | 0.80 | 3.18 | 0.62 | 5

SEED | 0.06 | 0.35 | 0.05 | 227

BCI-2b | 0.10 | 0.66 | 0.60 | 9

BCI-2a | 0.12 | 0.97 | 0.80 | 20

NER2015 | 0.21 | 1.45 | 1.20 | 4

The table is descriptive: it says that the displacement has a monotone component on most datasets, not that the component causes the loss. The main text keeps only the two quantities the argument uses, the within-cell partial correlation and the between-subject trend regressor.

## S26. Sleep telemetry: frozen-protocol replication

The protocol was written by the authors and fixed before the run (PROTOCOL_SLEEP_TELEMETRY.md); the package records the document, but its timestamp cannot be verified independently of the authors' record. Five stages with S3 and S4 merged, 0.5-30 Hz fourth-order zero-phase filter per series, 60-epoch series inside contiguous labelled runs, EEG Fpz-Cz and EEG Pz-Oz, log variance in four bands, LDA (lsqr, shrinkage auto) on the source series, completed-block mean with blocks of 15 and the first block unscored, lambda = 1 against lambda = 0, nights equally weighted within subject and subjects equally weighted.

Result: 44 recordings from 22 subjects, 3,945 scored cells. Subject-level difference -0.0483 ([-0.0588, -0.0369]), 21 of 22 subjects negative (two-sided sign test p = 1.1e-5), 30 of 44 recordings negative. Subject values in recording order: -0.027, -0.035, -0.054, -0.042, -0.061, -0.073, -0.056, -0.021, +0.025, -0.064, -0.103, -0.047, -0.038, -0.076, -0.052, -0.055, -0.066, -0.033, -0.036, -0.090, -0.025, -0.033.

The pre-declared interpretation applies: the sign replicates under a different acquisition protocol, at about half the magnitude, so the magnitude depends on the recording structure and the sign does not. The telemetry recordings belong to the same corpus as the discovery data and are not a new cohort; they are reported as a replication under a different protocol, not as independent external validation.

## S27. Decomposition of the sleep-EDF displacement

The analysis keeps the original task. The classifier is trained on the full source series, the scored target trials are the original ones (all 4,983 cells, first block of 15 unscored), and only the displacement that is removed changes. The displacement is written mu_B - mu_A = prior + conditional + unmatched, where prior is sum over shared classes of (pi_Bk - pi_Ak) mu_Ak, conditional is sum over shared classes of pi_Bk (mu_Bk - mu_Ak), and unmatched collects the classes present on only one side (3220 cells have a class only in the target, 3211 a class only in the source). The joint class-mix term is prior + unmatched. Reconstruction error: median 1.4e-14, maximum 2.1e-13.

quantity | value

uncorrected BA | 0.4415

full displacement removed | -0.098, 32 of 32 subjects negative

joint class-mix term removed (prior + unmatched) | -0.091, 32 of 32 subjects negative

shared-class prior only | -0.120, 32 of 32 subjects negative

class-conditional only | +0.034, 1 of 32 subjects negative

projected norms: prior / conditional / unmatched / total | 113.5 / 5.7 / 115.4 / 13.6

Reading. The joint class-mix term reproduces the loss almost exactly (-0.091 against -0.098 for the full displacement), while the conditional term alone gains. That joint test is the one that matters: the two mix components nearly cancel (113.5 and 115.4 against 13.6), so a prior-only correction is a much larger perturbation than the real one, and on its own could not settle the question. On the 693 cells whose class sets are identical the four values are -0.051 (full), -0.049 (joint mix), -0.049 (prior) and +0.031 (conditional).

Non-additivity. The three effects do not sum: the full displacement's effect minus the sum of the joint-mix and conditional effects is -0.041 ([-0.048, -0.034], 10,000 subject-level bootstrap), negative in 31 of 32 subjects. Balanced accuracy is a nonlinear functional of the decision scores, so a component that helps on its own need not protect when the other

component is present. The number is descriptive: it is computed from the same post-hoc displacements and is not an identified causal interaction.

Correction of the earlier version. A previous version restricted the cells to those whose class sets were identical, refitted the classifier on the retained classes and scored only the retained samples: 3,099 cells with an uncorrected accuracy of 0.654, not the task's 0.4415. That analysis measured a different task and is superseded.

## S28. Decoder and film-disjoint robustness

Sleep-EDF, every third cell (1,660 cells), same features, same warm-up, same scored trials. The correction is applied in feature space, x - lambda (mu_hat - mu_A), so it is defined for any decoder; features are standardised on the source series before fitting.

decoder | subject-level delta | 95 per cent interval | subjects negative

LDA (lsqr, shrinkage auto) | -0.105 | [-0.121, -0.090] | 32 of 32

RBF support-vector machine | -0.064 | [-0.074, -0.053] | 31 of 32

multilayer perceptron, 16 units | -0.061 | [-0.074, -0.049] | 30 of 32

SEED, film-disjoint. Fifteen films, five per class. One film of each class is held out, the classifier is trained on the remaining twelve and evaluated on the held-out three with the same correction; the procedure is repeated over five rotations, so every film is held out exactly once and the result is averaged within cell. Subject-level gain +0.189 ([+0.162, +0.218]), positive in 15 of 15 subjects, against +0.214 on the full film set. The three-rotation value of +0.200 and the single-rotation value of +0.147 are superseded.

Both checks use the same protocol as the main analysis and were written before the runs. Neither changes the sign of the main result; the nonlinear decoders and the held-out films both reduce its magnitude.

## S29. A label-free risk gate, and why it fails

Four label-free statistics were computed per cell on sleep-EDF (4,983 cells): the fraction of scored predictions changed by the correction, the mean absolute margin change, the mean displacement norm in decision space, and the disagreement between the completed-block and exponential means. Subjects were split 16/16; a threshold for each statistic was chosen on the development half to maximise the mean gain of a rule that corrects only below the threshold, then applied unchanged to the held-out half.

statistic | development threshold | held-out mean | cells corrected

prediction-flip fraction | 0.044 | -0.0011 | 9.4 per cent

mean margin change | 2.041 | -0.0022 | 12.8 per cent

displacement norm | 2.719 | -0.0024 | 13.2 per cent

estimator disagreement | 0.237 | -0.0082 | 10.6 per cent

always correct (reference) | - | -0.0963 | 100 per cent

Strictly online version. The statistics above are aggregated over the whole target sequence, so they are label-free but not causal: a deployment cannot know the flip fraction of a sequence before it is seen. The rule was therefore rerun block by block, deciding for each scored block from the trials and completed blocks that precede it, with the same threshold and no target labels. On sleep-EDF it corrects 10 per cent of blocks and returns +0.0001 against -0.102 for always correcting (16 of 32 subjects negative); applied unchanged to SEED it corrects 1 per cent of blocks and returns +0.0003, keeping none of the +0.214 over 45 cells. The offline version is superseded by this one, and the conclusion is the same: the gate buys safety by refusing to correct.

## S30. SEED estimator comparison under one protocol

Five causal estimators applied to the same 45 SEED session pairs with the same classifier, the same warm-up block of 15 segments and the same scored trials. Delta is balanced accuracy at lambda = 1 minus lambda = 0, averaged over cells.

estimator | delta | cells positive

completed-block mean | +0.2142 | 42 of 45

cumulative mean | +0.2136 | 42 of 45

variance ratio | +0.2164 | 42 of 45

sliding window 50 | -0.0408 | 22 of 45

exponential 0.05 | -0.0389 | 22 of 45

The long-memory estimates and the two short-memory rules differ by about 0.25 on the same cells, so on SEED the estimator is not a second-order choice: the gain exists only if the estimate has long memory. The same five rules all lose on sleep-EDF (S24), so estimator and task interact rather than one ranking holding everywhere.

## S31. Memory scale, block-wise decomposition and the SEED window sweep

Block-wise decomposition (the historical boundary preserved). The decomposition was rerun at block level: each scored block is corrected with the mean of the completed blocks that precede it, and that mean is split into its joint class-mix and class-conditional parts before projection. The split uses the union of source and target classes, so classes present only in the target contribute their unmatched term; per block, the two parts reconstruct the removed displacement to 1.7e-13 (median 1.0e-14).

correction applied | delta | subjects negative

full (class-mix + conditional) | -0.102 | 32 of 32

joint class-mix part only | -0.092 | 32 of 32

class-conditional part only | +0.020 | 3 of 32

The causal version reproduces the global picture almost exactly: -0.092 against -0.091 for the global diagnostic of S27, and +0.020 against +0.034 for the conditional part. The decomposition therefore does not depend on using a single global displacement. An earlier version of this section reported -0.126 for the mix-only correction; that run restricted the target classes to those the classifier had seen, so it dropped the unmatched term for the 3,220 cells whose target contains a class absent from the source. The per-block reconstruction check above fails for that version and passes for this one.

SEED memory sweep. Same classifier, same warm-up, same scored trials; only the memory of the estimate changes. Segments are 1 s, so the window in segments is the window in seconds.

window (segments) | 5 | 10 | 25 | 50 | 100 | 200 | 400

delta | -0.113 | -0.108 | -0.080 | -0.041 | +0.020 | +0.107 | +0.203

subject-level 95 per cent interval | [-0.145, -0.081] | [-0.141, -0.075] | [-0.114, -0.046] | [-0.075, -0.007] | [-0.015, +0.055] | [+0.069, +0.144] | [+0.164, +0.244]

exponential alpha (primed on the warm-up block) | 0.20 | 0.10 | 0.05 | 0.02 | 0.01

delta | -0.105 | -0.083 | -0.042 | +0.048 | +0.130

The effect is monotone in memory and changes sign between 50 and 100 s. A SEED film lasts 235 s: the median run of the label sequence is 235 segments in every session, because each film is one class. The crossover therefore sits at a quarter to a half of a class episode, as the memory-scale account predicts. No stage-duration statistic is used to predict the sleep-EDF sign: the whole-night run-length distribution does not describe the windows that were actually scored, and the per-block composition of those windows is reported separately below. The two datasets are therefore left with two separately established failure conditions rather than one shared mechanism.

Class composition, all blocks. Of the 14,949 scored blocks, 8,301 (55.5 per cent) are dropped by the two conditions the balanced-accuracy analysis needs: 6,446 have a single-class scored block, 2,970 a single-class history window, and 1,115 satisfy both, leaving 6,648 blocks (44.5 per cent). Dropping them removes the most concentrated cases first, so a claim about concentration cannot rest on that subset. Two analyses are reported. (i) Within the 6,648 blocks that remain, the paired subject-level difference between a high (0.8 or more) and a low (below 0.6) maximum class fraction is -0.009 ([-0.029, +0.012], 13 of 31 subjects, two-sided sign test p = 0.47): no association. (ii) Counting every block and using, instead of balanced accuracy, the change in the fraction of misclassified epochs, the same paired difference is +0.084 ([+0.057, +0.109]), 29 of 32 subjects, two-sided sign test p = 2.6e-06 and one-sided p = 1.3e-06 in the predicted direction: a history dominated by one class goes

with 8.4 percentage points more errors. The sign convention is stated because the test is directional. Block-level Spearman correlations over all 14,949 blocks (+0.12 for the maximum class fraction, -0.13 for entropy, ties corrected) are descriptive only: blocks within a subject are not independent observations, so the inference rests on the subject-level pairs. This remains an exploratory, label-assisted diagnostic; it does not show that composition causes the loss, only that the two co-vary within subjects.

## S32. Withdrawn: the class-sorted cache analysis

An earlier version of this work computed the grasp-and-lift results on a feature cache whose rows were sorted by class label rather than left in acquisition order. The grasp-and-lift numbers below were produced that way and are withdrawn as evidence about the recordings; they are kept because the audit of how the error arose is part of the paper (Section 4.2). The other datasets in Table S8 - SEED, BCI-2a, BCI-2b and NER2015 - were computed on caches that were never sorted and are unaffected by this error, but they belong to the same development analysis and are not among the primary results of this paper.

Withdrawn values on grasp-and-lift: cumulative mean -0.115, variance-ratio shrinkage -0.095, predictive shrinkage -0.011 (S1); a label-free gate at -0.047 (S1, S29); a temporal-structure association of rho = +0.776, with the prior and conditional block terms at +1.27 and -0.27 of the observed movement and a long-short interaction of +0.031 (S21).

Current values on the same data in acquisition order: +0.001 (cumulative mean), +0.004 (variance ratio), +0.006 (predictive shrinkage). There is no correction loss to explain, so there is nothing for a trend association or a gate to predict (Section 4.2 and Section 5).

What the error teaches. A cache sorted by class turns the block structure into a staircase of class means: within a block the trials belong to one class, across blocks the class changes, so the past-block mean estimates a class mean rather than a recording offset and subtracting it removes class information. The SEED memory sweep (S31) exposes a failure with the same shape at short memory. The two share a risk profile - an estimate whose memory is short relative to the class structure removes class information - but they are not asserted to share a cause: one is a data-handling artefact, removed by fixing the order of the data, and the other a property of the estimator itself.

Moved here from S21: the temporal-structure analysis of the block displacement and the decomposition that accompanied it. Table S8 covers five datasets; only its grasp-and-lift column was computed on the cache sorted by class, and only that column is withdrawn. The SEED, BCI-2a, BCI-2b and NER2015 columns were computed on caches that were never sorted and are unaffected by this error, though their reading is still governed by the same caution as the rest of S21 - they describe a development analysis, not the primary results of this paper.

If the right strength is a property of the recording, it should be visible in how the target mean moves. For every cell we took the block displacements $d_b = P W (\mu_b - \mu_A)$, fitted a

linear ramp to each score component, and measured $R^2$, the share of their variance the ramp explains. The number is only interpretable against a null: with five blocks, block order alone produces $R^2 \approx 0.25$ by chance. We therefore permuted the block order 500 times per cell and report the excess $\Delta R^2 = R^2 - E[R^2_null]$.

**Table S8. Temporal structure of decision-space displacements and its relation to the full-strength correction loss, per dataset, over five-block summaries.**

| dataset | blocks | R2 | R2 under permuted order | excess dR2 | lag-1 | loss |
|---|---|---|---|---|---|---|
| grasp-and-lift | 5 | 0.800 | 0.255 | +0.528 | +0.65 | +0.102 |
| SEED | 227 | 0.063 | 0.004 | +0.059 | +0.95 | −0.181 |
| BCI-2b | 9 | 0.104 | 0.130 | −0.017 | −0.16 | −0.020 |
| BCI-2a | 20 | 0.120 | 0.053 | +0.067 | +0.17 | −0.024 |
| NER2015 | 4 | 0.213 | 0.329 | −0.096 | −0.39 | 0.000 |

Across the twelve grasp-and-lift subjects this analysis gave rho = +0.776 (p = 0.003), it survived using only the four equal-length blocks (rho = 0.706) and leaving out any single subject (0.727 to 0.864), while within the other datasets it was weaker (SEED 0.364, BCI-2b 0.200, BCI-2a 0.450, NER2015 -0.035). The grasp-and-lift column of this table, and the decomposition in the paragraphs below, were computed on the cache sorted by class; the other columns were not. In acquisition order grasp-and-lift shows no correction loss for a trend to explain, so the grasp-and-lift part of this analysis is withdrawn as evidence about recordings and kept only as part of the audit.

What the block mean contained, on the sorted cache. The oracle result above ruled out lag as the operative mechanism, and a block-level decomposition said what replaced it. Writing the block mean as mu_b = sum_k pi_bk mu_bk, the displacement splits exactly into a class-prior term sum_k (pi_bk - pi_sk) mu_sk and a class-conditional term sum_k pi_bk (mu_bk - mu_sk), and the reconstruction is exact to machine precision (median relative error 1e-16). Projected on the discriminant directions, the prior term accounted for a median of +1.27 of the observed movement and the conditional term for -0.27. In that analysis the separation was stable across block lengths: at m = 10, 15 and 20 the block-prior correction gave 0.358, 0.382 and 0.510 against 0.773, 0.773 and 0.770 uncorrected, while the class-conditional correction gave 0.851, 0.836 and 0.811. These are historical numbers from the sorted cache: they describe how the artefact behaved, not what happens in the recordings, and no claim in this paper rests on them.

Interaction, stated explicitly. With long memory taken as the mean of the cumulative and completed-block estimators and short memory as the mean of the sliding-window and exponential ones, the interaction is the paired within-subject difference of their full-strength changes: +0.031 [+0.029, +0.034] on grasp-and-lift and +0.255 [+0.205, +0.306] on SEED. The grasp-and-lift value is small but not zero, and it is smaller than the spread between the worst and the best estimator there (0.108 to 0.163); the SEED value is larger than any estimator effect in the paper.

Three limits belong with it. The permutation null is iid, so it destroys the temporal correlation that the block means actually carry (lag-1 is 0.65 on grasp-and-lift and 0.95 on SEED); $\Delta R^2$ therefore measures order dependence, not specifically a linear drift, and a block-bootstrap null would be more conservative. Five blocks give the $R^2$ a coarse resolution, which is why the inference is at the subject level and not per cell. And the association is an association: the trend covaries with the loss, it does not identify what produced the trend. As a check that does not destroy the serial dependence, we also compared each cell with the circular shifts of its own block sequence ($B - 1$ shifts per cell, wrap-around kept). The observed median $R^2$ is 0.800 against 0.236 under circular shift, an excess of 0.567, and 92.9 per cent of the 336 cells lie above their own circular null. The circular null preserves the within-session correlation but introduces an artificial wrap discontinuity, so it is a sensitivity check rather than a definitive test.

## S33. Two controls for the class-mix reading

Two controls were run against the reading that the sleep loss is carried by the class-mix displacement: one that holds the structure of the history fixed, and one that removes the prior term using proportions estimated without labels.

Structure matching, done within subject and within stratum. An earlier draft of this section reported a matched concentration contrast of +0.050 (19 of 28 subjects, sign test $p = 0.087$). That number has been withdrawn: a full rerun recording per-block class concentration, with five definitions of the high and low arms and at least two blocks per arm inside every subject-by-stratum cell, does not reproduce it. The median split gives -0.023 (10 of 32 subjects positive, sign $p = 0.050$), the 0.6 and 0.7 thresholds give -0.006 and +0.005, the tercile split -0.004, and the uniform-share split is usable in only three cells; none is significant and none is positive at the reported size. The conclusion is that no reproducible block-level association between class concentration and the size of the correction effect was found, which is consistent with the main result holding across concentrations rather than being driven by them.

Label-free prior estimate, on the cells where the estimator's assumptions hold. BBSE estimates the target class proportions from the source classifier's predictions on the unlabeled target series; it requires every target class to have support in the source, which holds for 1,763 of the 4,983 cells (35.4 per cent). The other 3,220 cells, whose target series contains a stage absent from the source, are excluded from this estimate and reported separately. On the valid cells the confusion matrix is well conditioned (median condition number 2.0) and the estimated proportions have a median total-variation error of 0.207 over the full class space (0.285 on the cells with unseen classes, where the missing mass is counted). Removing the displacement built from the estimated proportions costs -0.078 ([-0.096, -0.060], 30 of 32 subjects negative); removing the displacement built from the true proportions, on the same cells and with the same normalisation, costs -0.110 ([-0.127, -0.092], 32 of 32). Both are harmful and the estimate is the milder of the two, so the estimation error attenuates the damage rather than removing it. The estimate uses the whole target series, so it is transductive and not a past-only rule.

What these two controls change. They bound the class-mix reading rather than confirm it. Structure matching inside each subject does not produce a reproducible positive association: the original +0.050 could not be regenerated under five definitions of the arms, and the regenerated estimates lie between -0.023 and +0.005 with no significant effect (Section S33). The label-free estimate of the class mix can be computed only on the third of the cells where every target class exists in the source. Neither control turns the displacement into something a deployment could use, and BBSE does not establish that a change in class composition is the cause of the loss.

Tables from earlier versions of this supplement are not reproduced here. The numbers they reported are in the main text (Tables 1-8) and in the analysis archive: results/master_cells.csv, results/final_compare.txt, results/lambda_control.txt and results/phase30.txt.

## S34. Material moved from the main text

This section holds analyses that the main text now summarises in one or two sentences. Nothing here is new: it is the text removed during the revision that added Sections 4.6 to 4.8, kept so that the numbers cited in the main text remain checkable.

Removed text, by original paragraph index.

[[108]] Estimator memory on SEED. The five estimators compared in Section 4.2 were rerun on SEED under one protocol - same warm-up block, same scored trials, same classifier. The completed-block mean gains 0.214 (42 of 45 cells positive), the cumulative mean 0.214 and the variance-ratio rule 0.216, while the 50-trial window loses 0.041 and the exponential average 0.039. The gain therefore does not come from correction as such: it needs an estimate with long memory, and the two short-memory rules remove it entirely. On sleep-EDF the same five rules all lose (Section 4.7). The estimator and the task therefore interact, and the paper reports the interaction rather than a ranking of estimators.

[[109]] Memory scale. The estimator comparison suggests that the relevant quantity is the memory of the estimate relative to the duration of a class episode, so the memory was swept with everything else fixed. On SEED, averaging over the last 5 scored segments gives -0.113, over 10 -0.108, 25 -0.080, 50 -0.041, 100 +0.020, 200 +0.107 and 400 +0.203; an exponential average moves the same way with its rate (-0.105 at alpha = 0.2, +0.130 at alpha = 0.01, with the exponential average primed on the warm-up block so that all estimators start from the same history). Subject-level intervals: -0.113 [-0.145, -0.081] at 5 segments, +0.020 [-0.015, +0.055] at 100 and +0.203 [+0.164, +0.244] at 400, with 1, 8 and 15 of 15 subjects positive. The within-subject slope of the effect against log window is positive in all 15 subjects (mean +0.071 per log unit). The effect is monotone in memory and changes sign between 50 and 100 s, while a SEED film lasts 235 s (median run of the label sequence, identical across sessions): the crossover sits at about a quarter to a half of a film, as the memory-scale account predicts. The account is not extended to sleep-EDF: there the loss follows the class-mix part of the displacement, and no stage-duration statistic is

used to predict what a memory sweep would do. Whether the two cases share a mechanism is left open rather than asserted.

[[134]] Where the sleep loss comes from. The displacement of the target mean splits exactly into a class-mix part (shared-class prior plus the classes present on only one side) and a within-class part, and the split can be applied without touching the task: the classifier is the one trained on the full source series, the scored trials are the original ones, and only the displacement that is removed changes (reconstruction error 2.1e-13 over all 4,983 cells). Removing the whole displacement costs 0.098; removing only the joint class-mix part costs 0.091 (32 of 32 subjects negative); removing only the shared-class prior alone costs 0.120; and removing only the class-conditional component gains 0.034, with 1 of 32 subjects negative. The joint test matters because the two mix components nearly cancel - projected norms 113.5 and 115.4 against a total of 13.6, with the conditional term at 5.7 - so a prior-only correction would have been a much larger perturbation than the real one. The joint term reproduces the loss, which is what the class-mix reading predicts; on the 693 cells whose class sets are identical the same four numbers are -0.051, -0.049, -0.049 and +0.031. The analysis locates the harm; it does not explain why the proportions move, and it uses target labels and a single global displacement, so it is a diagnostic, not an online method. The two components do not add: on the same cells the non-additive difference, defined as the full displacement's effect minus the sum of the two component effects, is -0.041 ([-0.048, -0.034]), negative in 31 of 32 subjects. Balanced accuracy is a nonlinear functional of the scores, so the two shifts do not act independently - removing the conditional part alone helps, yet removing it alongside the mix part does not protect against the loss.

[[154]] The cause is identified only in part. On the original task the decomposition of Section 4.7 shows that the harm travels with the joint class-mix part of the displacement: removing the within-class component alone gains 0.034, removing the joint class-mix part loses 0.091, and removing the shared-class prior alone loses 0.120 - the larger prior-only number is a perturbation the real displacement does not contain. On SEED the sign is monotone in the memory of the estimate and changes between 50 and 100 s, while films last two to four minutes; the same crossover does not occur on sleep-EDF, where every estimator loses. Because the mix components are an order of magnitude larger than their residual, the analysis says which part carries the loss, not why the proportions move. As for a label-free rule, none of the rules tested here - the gate fitted on one half of the subjects, the amplitude and dispersion statistics, or the sign models - held up across tasks: the gate avoids the sleep loss only by refusing to correct, and then discards the entire SEED gain. We state this as the result of the rules we tested, not as a claim that no such rule can exist. The model here is explanatory, not general: a symmetric equal-variance Gaussian with a linear threshold explains why a composition-only correction must hurt and how much drift is needed to reverse it, and the certificate is an identity for a fixed affine decoder. Neither is offered as a theory of domain adaptation, and the paper does not claim that the sign of the correction is predictable in general.

[[79]] The two tables in this subsection use different cell sets — Table 4 includes the two target-informed representations and Table 5 does not — which is why the BCI-2a baseline differs between them; Section 3 gives the counts.

[[85]] We compare four estimators of mu_hat: a cumulative mean, a sliding window of 50 trials, an exponential average with alpha = 0.05, and a shrunk cumulative mean with tau = 50. Table 5 reports the primary comparison on source-only representations with a common online protocol; Table 4 additionally includes the two target-informed representations and is therefore not a strictly causal comparison.

[[87]] The pattern, not the ranking, is what matters: the benchmark that gains most and the benchmark that loses most respond to the same knob in opposite directions (Section 4.4). Three regimes appear - a constant offset is corrected by anything, drift punishes long memory (0.08 to 0.10 over the cumulative mean, 0.14 for ADWIN), and a change of class proportions costs the block rules 0.01 to 0.02 while leaving the shrinkage rules almost unchanged.

[[92]] Once the window is fixed the comparison is unambiguous on SEED: the variance-ratio rule keeps +0.211 of the +0.250 that the transductive correction achieves there, gains about 0.03 on the two motor-imagery datasets and is flat on NER2015. On the acquisition-order grasp-and-lift cache no rule loses: cumulative +0.001, window +0.002, exponential +0.002, block mean +0.001, variance-ratio +0.004, predictive +0.006. The predictive rule's leave-one-block-out weight and the cold-start fallback are derived in Supplementary Section S23 and implemented in the repository's gal_estimators_time.py, which produces the numbers above.

[[93]] Block length was fixed at m = 15 because grasp-and-lift sessions hold 28-34 trials and sleep-EDF series 60 epochs; at m = 25 two blocks of history remain, at m = 50 one.

[[95]] The comparisons above are observational: they say which estimator wins where, not why. A controlled generator - constant offset, linear drift, step changes at block boundaries, or class-proportion change - reproduces the qualitative pattern with no uniformly superior rule. A constant offset is repaired by every rule (0.500 to about 0.70). Under drift all rules improve on no correction, and the adaptive window gains most (0.519 to 0.663 against 0.617 for the cumulative mean and 0.602 for the variance-ratio rule). Under step changes the ordering is the same and the spread is larger (0.342 to 0.607 for ADWIN, 0.440 for the cumulative mean). Under class-prior change every corrected rule sits slightly below no correction (0.704 against 0.688-0.698). The simulations are hypotheses about mechanisms, not causes, and they do not identify which mechanism dominates a recording.

[[59]] Which tests are primary. Five tests were pre-specified as one family with Holm control: the subject-level Wilcoxon signed-rank comparison of mean against none, per dataset. After Holm the p-values are 3.05e-4 (SEED), 0.016 (BCI-2a), 0.016 (BCI-2b), 0.104 (grasp-and-lift) and 0.320 (NER2015), so the gain survives correction on three datasets, not five. The family is pre-specified, but it is not five independent confirmations of one causal

claim, and this paper does not present it that way: SEED is estimated on the released DE features whose preprocessing we did not control, BCI-2a and BCI-2b were already used during development, and the two remaining datasets fail after correction. Section 4.4 analyses one BCI endpoint under a fixed protocol with a causal filter and a source-only alignment, and that endpoint is reported separately as prospective-in-explored-data rather than folded into a single confirmed chain. Everything outside the five-test family is exploratory without a multiplicity correction, and per-dataset numbers are reported with their intervals throughout.

[[60]] Preprocessing detail. The 2a/2b features use a fourth-order Butterworth 8-30 Hz band-pass on the continuous signal. BCI-2a is epoched from 2.0 to 6.0 s after the trial onset, which is 0.0 to 4.0 s after the cue (the cue follows the trial marker by 2 s in that dataset, per the official data-set description [26]); BCI-2b is epoched from 0.5 to 3.5 s after the cue, which is 3.5 to 6.5 s after the trial marker (there the cue follows the marker by 3 s, measured from the GDF annotations). The two windows are therefore not the same, and this is a protocol difference between the datasets rather than a common setting; euc is a per-trial log-variance in four consecutive time windows of the epoch; csp takes the generalised eigenvectors of the first class covariance against the average covariance, three filters per class, fitted on the training session; tan maps the session covariance to the tangent space; euc_ea divides by the session covariance. LDA uses solver=lsqr and shrinkage=auto.

[[95]]

[[155]] Matching on class composition does not change this picture. When blocks are compared only within strata that share the block class set and the history class set, and further within a 0.1 prior bin, the matched difference is -0.104 with none of the 32 subjects positive (six specifications, 734-930 strata, all within 0.004 of each other). What this shows is robustness of the negative effect to how the strata are defined: it does not show that class composition is not the mechanism, because the composition of a block and its history is exactly the quantity being held fixed, and the oracle decomposition of Section 4.6 already points at the mix component.

[[156]] An independent negative control points the same way. On a P300 speller dataset that took no part in development (16 subjects, single session, target prior 0.14), the source is the first half of each subject's epochs and the target the second half, so the two share their class composition and no drift separates them. The pre-registered prediction was that the correction would hurt; it did not. The measured effect is -0.0005 with a 95 per cent interval of [-0.0024, +0.0012] and 7 of 16 subjects positive. The interval lies inside an equivalence margin of 0.02 on balanced accuracy, but the pre-registered run decoded at chance and the exploratory run only reached 0.50-0.78, so we report that no reliable gain or loss was observed under this protocol and we do not claim that the effect is zero. The diagnosis we do offer is narrower: with source and target sharing their class composition, the displacement the correction removes is approximately zero, which is the branch of the

theory that predicts no change. The prediction was wrong about the sign, the diagnosis explains why, and both are reported.

[[157]] A second independent corpus has the cross-session structure this paper is about and gives a third answer. The BCI-challenge error-potential data (16 subjects, five sessions each, four consecutive forward pairs) give an overall mean change of -0.0044 with a 95 per cent interval of [-0.0166, +0.0075] and 6 of 16 subjects positive. Conditioning on decoding quality changes the picture: in the six subjects with balanced accuracy at least 0.52 the mean change is -0.0239 [-0.0415, -0.0093], and the per-subject correlation between balanced accuracy and the correction effect is -0.71 ($p < 0.01$). That subset was chosen after seeing the data and is exploratory, not a replication; what it does show is that the near-zero average is not evidence that the correction is harmless where the decoder works.

## S35. Estimator comparison tables moved from the main text

Tables 4 and 5 of earlier versions are kept here. The main text now reports the pattern and the controlled comparisons of Sections 4.4, 4.6 and 4.7, and points to this section for the full ranking.

Table 4. Balanced accuracy on the common post-prefix test set (trials k+1 … n; k = 20, or 50 for SEED), LDA, all representations including the two target-informed ones; these cells are transductive and are not the strictly causal comparison, which is Table 5. batch is the transductive correction scored on the same trials.

Table 5. Causal (past-only) mean estimators evaluated on the common post-warm-up test set (trials 16–n, m = 15), using source-only feature representations. The first 15 trials are excluded from balanced-accuracy evaluation. block mean is the unweighted mean of completed blocks; Eq. (9) applies variance-ratio shrinkage to that mean; predictive uses leave-one-block-out shrinkage. The last column reports the subject-level paired difference between Eq. (9) and no correction.

## S36. Methods for the three mechanism experiments

This section documents the analyses added to Sections 4.6 to 4.8. All code and result files are in the release described under Data availability, and every number quoted in the main text can be recomputed from them.

### Online decomposition (Section 4.6).

Protocol. For each of the 4,983 sleep-EDF cells the classifier is the one trained on the earlier series of the same night (LDA, lsqr solver, automatic shrinkage), the scored epochs are epochs 16 to 60 of the later series with no exception, and the estimator is the completed-block mean of 15-epoch blocks, so the correction for block k uses blocks 1 to k-1 only. Three decompositions are computed on identical history: the anchored one used in the paper, the unanchored variant retained as a diagnostic, and the reconstruction residual.

Definitions. pi_A,c is the class proportion of the source series, m_A,c its class-conditional mean, mu_A its overall mean, pi_hat_c and m_hat_c are the block-mean-weighted proportion

and mean over the completed blocks, and u is the total mass of classes present in the history but absent from the source. The anchored terms are sum_{c in A}(pi_hat_c - pi_A,c)(m_A,c - mu_A), sum_{c in A} pi_hat_c (m_hat_c - m_A,c) and sum_{c not in A} pi_hat_c (m_hat_c - mu_A).

Checks. Reconstruction error is reported per block and per cell; the maximum over 4,983 cells and 14,949 scored blocks is 2.7e-14. Origin invariance is tested in two ways: analytically (an unmatched class of mass u moves the unanchored terms by -u a and +u a under a translation a) and numerically, by translating every feature by four units with an unmatched class present and confirming that the anchored terms move by less than 1e-9 while the unanchored ones move by more than 1e-3. The test is part of the released test suite.

## Factorial intervention (Section 4.7).

Generator. Class-conditional means and a pooled covariance are estimated per subject from the real sleep-EDF features of the first available night (8 log band variances, two channels). Source: 600 epochs, classes balanced by construction. Target: 600 epochs drawn from the same Gaussians with a controlled class proportion; the composition intervention adds 0.25 of mass to one class and renormalises, the drift intervention adds a constant vector of magnitude drift * sqrt(trace(cov)/d)/sqrt(d) in a random direction. Decoder, block length, warm-up and correction are those of the main experiment.

Design and inference. The main factorial is 12 subjects x 20 seeds x 2 levels of the composition factor x 3 levels of the drift factor = 1,440 independent data streams. The five-strength sweep draws 12 x 8 x 2 x 3 = 576 streams, evaluates five strengths on each and repeats every evaluation twice, which is 5,760 strength evaluations summarised as 2,880 strength-condition rows (two repetitions averaged per row). The dense grid draws 12 x 6 x 2 x 2 = 288 streams and evaluates eleven strengths once on each, giving 3,168 rows; each row is one stream at one strength. Repeats are averaged inside a subject, and subjects are the inference unit: the sample is 12, not 1,440, 2,880 or 3,168.

Optimum. The closed form lambda* = b/(b+c) is a prediction of the one-dimensional symmetric model. In the five-class, multi-dimensional generator it is used as a heuristic for the direction of the optimum; the measured optimum is located on the grid and reported with the grid resolution, not with a precision the grid cannot support.

## Memory sweep (Section 4.4).

SEED: the archived per-cell values of the window and exponential sweeps are aggregated per subject and bootstrapped; segment length is one second because the features are distributed 1 s differential entropy. sleep-EDF: the same 4,983 cells are re-scored with the correction limited to the last one, two or three blocks of history, everything else unchanged. A series holds four blocks, so three is the maximum and equals the full-history case.

Boundaries. The factorial generator is Gaussian with a pooled covariance, so it establishes feasibility of the mechanism and the scope of the theory; it is not a physiological model and does not identify the cause of the stage-proportion change in the real corpus.

## S37. Proofs of Propositions 1-4

Setting. The decoder is affine, s_k(x) = w_k'x + b_k, with prediction argmax_k s_k(x). Write W for the K x D weight matrix and F = I - 11'/K for the centring matrix, so that F W d = 0 iff every pairwise score difference is unchanged by a displacement d. The three pipelines are none (no correction), mean (fit on X_A - mu_A, apply to X_B - mu_B + mu_A) and zero (fit on X_A, apply to X_B - mu_B).

Proposition 1. Under x -> x + c on both sessions with the classifier refitted, none and mean return identical predictions and zero does not. Proof. For none, the fit on X_A + c keeps the weights and moves the intercepts to b_k - w_k'c; the test features move to x + c, so the score is w_k'(x + c) + b_k - w_k'c = w_k'x + b_k. For mean, the training mean becomes mu_A + c and the target mean mu_B + c, so (X_A + c) - (mu_A + c) = X_A - mu_A and (X_B + c) - (mu_B + c) + (mu_A + c) = X_B - mu_B + mu_A: identical features. For zero, the corrected target is (X_B + c) - (mu_B + c) = X_B - mu_B while the intercepts moved by -w_k'c, so the score changes by -w_k'c, which is zero for all k only if F W c = 0.

Proposition 2. A correction x -> x - gamma(X) is translation equivariant iff gamma(X + c1) = gamma(X) + c for all c. Proof. Immediate from the requirement that the corrected features x + c - gamma(X + c1) equal the untranslated corrected features x - gamma(X) for every x. The sample mean satisfies it because mean(X + c1) = mean(X) + c; the coordinate-wise median satisfies it for the same reason; a reference taken from the training session plus a constant satisfies it. The identity characterises equivariance, and any claim that only the sample mean has this property is false.

Proposition 3. A displacement d leaves every pairwise score difference unchanged for all inputs iff (w_j - w_k)'d = 0 for all j, k, i.e. F W d = 0. Proof. s_j(x - d) - s_k(x - d) = s_j(x) - s_k(x) - (w_j - w_k)'d, so the differences are unchanged for every x iff (w_j - w_k)'d = 0 for every pair. The set of such d is the null space of F W, a subspace of dimension at least D - (K - 1); it is not the span of the rows of W, which is a K-dimensional subspace of a different kind.

Proposition 4. Adding a constant a to the score of one class leaves every one-vs-rest AUC of that class unchanged. Proof. A one-vs-rest AUC compares the scores of positive samples with the scores of negative samples: [s_c(x+) + a] - [s_c(x-) + a] = s_c(x+) - s_c(x-), so every positive-negative comparison, and therefore the AUC, is unchanged. The statement holds within one score function; it does not extend to comparing raw scores with pairwise margins, which need not order samples alike.

| dataset | none | batch | cumulative | window | exponential | shrunk | one-sided patch |
|---|---|---|---|---|---|---|---|
| BCI-2b | 0.7063 | 0.7428 | 0.7417 | 0.7419 | 0.7420 | 0.7327 | 0.5839 |
| BCI-2a | 0.5640 | 0.6111 | 0.6077 | 0.6145 | 0.6138 | 0.6043 | 0.5169 |
| SEED | 0.4599 | 0.7094 | 0.6782 | 0.4194 | 0.4178 | 0.6845 | 0.3394 |
| grasp-and-lift | 0.787 | 0.797 | 0.789 | 0.789 | 0.789 | 0.791 | 0.519 |
| NER2015 | 0.5878 | 0.5883 | 0.5840 | 0.5842 | 0.5822 | 0.5867 | 0.5857 |

| dataset | none | cumulative | block mean | Eq. (9) | predictive | Eq. (9) − none |
|---|---|---|---|---|---|---|
| BCI-2b | 0.7062 | 0.7383 | 0.7385 | 0.7359 | 0.7332 | +0.030 |
| BCI-2a | 0.4970 | 0.5291 | 0.5261 | 0.5258 | 0.5291 | +0.029 |
| SEED | 0.4594 | 0.6730 | 0.6736 | 0.6706 | 0.6747 | +0.211 |
| grasp-and-lift | 0.787 | 0.789 | 0.789 | 0.791 | 0.793 | +0.004 |
| NER2015 | 0.5916 | 0.5873 | 0.5873 | 0.5888 | 0.5898 | −0.003 |